

<https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/avr-gcc>

GATE Question solution using Avr-gcc

Question

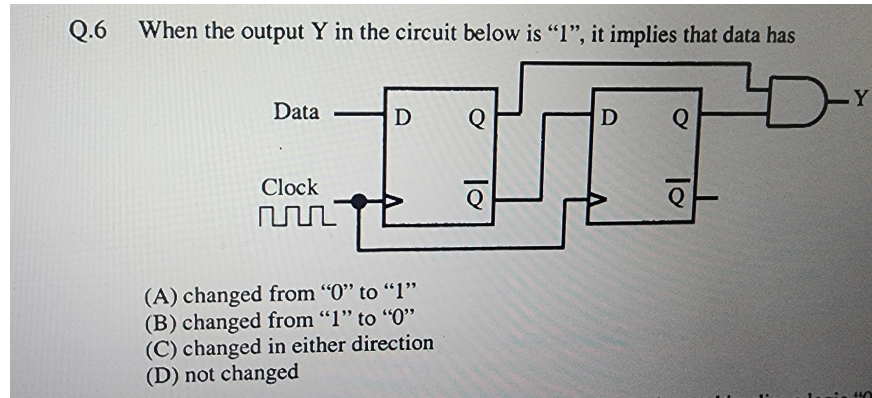


Figure 1: Original Q.6 from Exam Paper

Solution

To determine when output Y in the circuit is "1", analyze the edge-detection logic using two D flip-flops. Y is generated by the AND of the first flip-flop's Q output and the negation of the second flip-flop's Q output:

$$Y = Q_1 \cdot \overline{Q_2}$$

where Q_1 is the current data and Q_2 is the previous data.

Truth Table

Q ₂ (Previous)	Q ₁ (Current)	Y
0	0	0
0	1	1
1	0	0
1	1	0

K-Map Minimization

	Q ₂ = 0	Q ₂ = 1
Q ₁ = 0	0	0
Q ₁ = 1	1	0

Thus, the minimized Boolean expression:

$$Y = Q_1 \cdot \overline{Q_2}$$

Implementation Steps (AVRGCC on Termux)

1. Install Termux and the necessary GCC toolchain for AVR: `pkg install clang avrgcc avrdude`
2. Write the C code below for edge detection
3. Compile your code: `avr-gcc -mmcu=atmega328p -o edge.bin edge.c`
4. Flash to an AVR device using avrdude, or run with a simulator.

Conclusion

This circuit and corresponding software detect a rising edge on the data line, outputting “1” only when the data transitions from “0” to “1”. The function was minimized using a Karnaugh map to $Y = Q_1 \cdot \overline{Q_2}$. The provided C code and hardware description allow implementation with microcontrollers or logic ICs like 7474 in a FSM arrangement for precise edge detection.